Week 1: Learning the Tools

1. Introduction

The laboratories for this course are spread across two rooms: the Electronics Lab (Elec 210), and the Embedded Systems Lab, or ESL (Elec 204). For this course, the ESL acts as an overflow - fill up the electronics lab first! Philipp Hof, Randy Hampton and Diego Ramirez are the technicians in charge of these rooms. Please report any damage or other issues with equipment to one of them, or a TA.

Each bench in the electronics lab is fitted with an Agilent DSO-X 2012A model oscilloscope. A different (but similar enough) oscilloscope is present in the ESL. This, along with the workstation computer and the software on it, are the main tools you will use in this course.

You should also find a stack of PCBs and a USB-C cable on your desk. This is the device you'll be programming during the course. Leave it on your desk at the end of the lab; you will be issued one to use for your project next week.

2. Lab Procedure

All labs, including this one, are designed to be completed in pairs. Find a partner during this first session who is in the same lab stream as you. While this person doesn't *have* to be your project partner, it's not a bad idea. You need to have formed project groups and entered them into Learn by the end of Thursday 20th February. If you're not in a valid group by then, you will be assigned a partner randomly.

One member of each pair should log into the lab computer as soon as possible. Logging in can take a while, especially for the first time (i.e. this lab session specifically).

For each lab, complete as much of these handouts as you can within your allocated 2-hour lab session. Don't be despondent if you do not complete everything. You may also access the Electronics Lab at other times during normal working hours, Mon - Fri, provided it is not being used for other formal lab sessions.

3. The ST Nucleo & RCAP

The Nucleo is a self-contained development board manufactured by ST Microelectronics. Development boards typically provide an easy way for prototyping and evaluating a microcontroller (MCU) for the use in a project. The specific board you will be using for this project features the STM32C071RB Cortex-M0+ MCU, which supports a user-programmable push-button switch (B1), two user-programmable LEDs (LD1 & LD2), a number of peripherals (ADC, timers, etc.) and a USB interface. Fitted underneath the Nucleo is a daughterboard that extends its functionality. This board is designed in-house by Philipp Hof and is referred to as the Robust Collection of Additional Peripherals (RCAP). It supports a digital display, a joystick and an inertial measurement unit (IMU) that you will use in future weeks. Documentation on the Nucleo board, the STM32C071 MCU and the STM32Cube toolchain can be found on your course Learn pages.

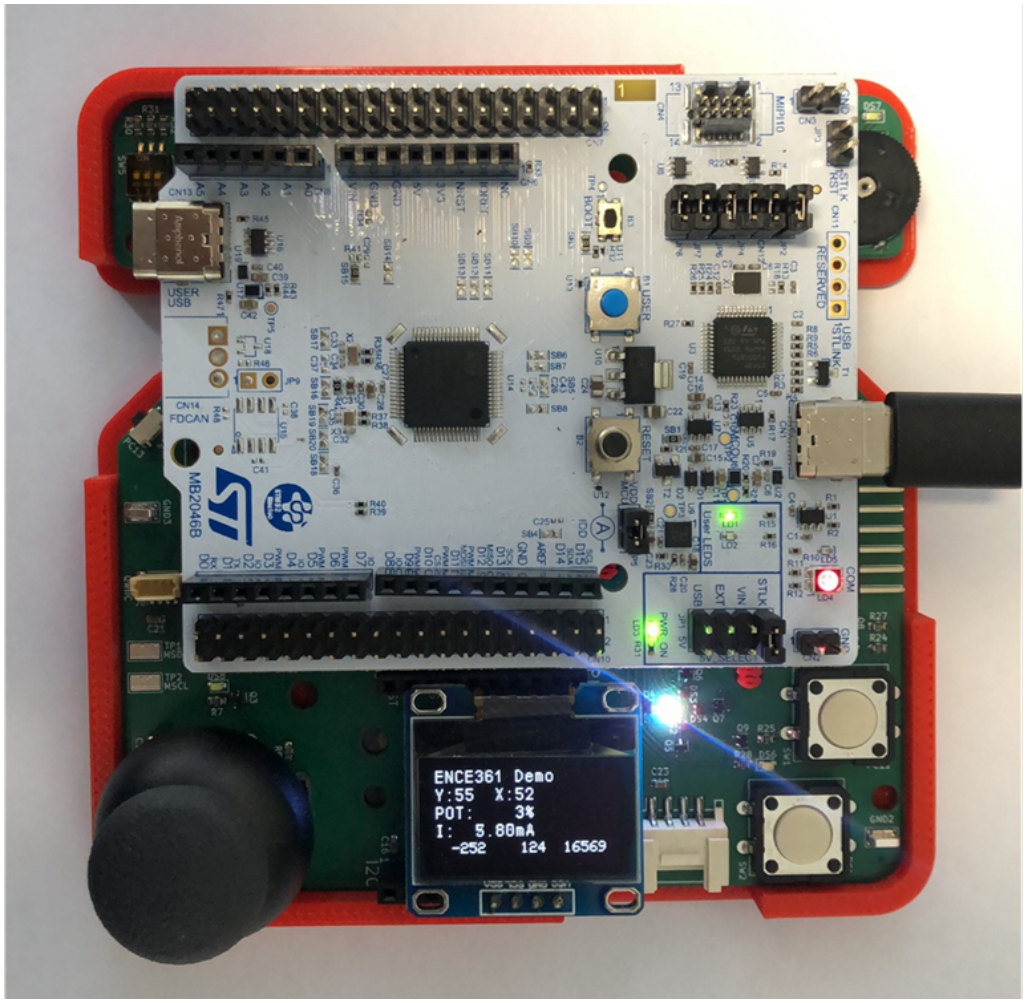

Figure 1: The Nucleo and RCAP boards, running a test program.

Plug your Nucleo board into the computer using the USB port shown above. The pre-loaded test program should begin running, and both boards should show some activity. Check:

- LD1 (Green LED on the Nucleo) blinks about twice a second.
- LD2 (Blue LED on the Nucleo) blinks quickly if SW3 or SW4 are pressed.
- The right-hand side RGB LEDs flash, and change colour as you move the joystick.
- Pressing SW1 sounds the buzzer.
- Pressing SW2 activates the haptic feedback.
- Pressing the joystick in sounds the buzzer at a different frequency.
- The OLED shows the joystick position, potentiometer value, current consumption, and the accelerometer data.

3. The Oscilloscope

An oscilloscope (or just "scope") can measure a range of parameters, such as the amplitude of a voltage signal, peak-to-peak voltage, frequency and duty cycle. They can measure and display multiple independent signals simultaneously. They are by far the most important tool, besides a computer, for working on embedded systems.

The electronics lab has Agilent DSO-X 2012A scopes. The ESL has Rhode & Schwartz RTB2004 scopes. This guide is mostly based around the electronics lab scopes, but will cover the use of both. Because you won't always have access to the same model of scope, you should focus on learning general use.

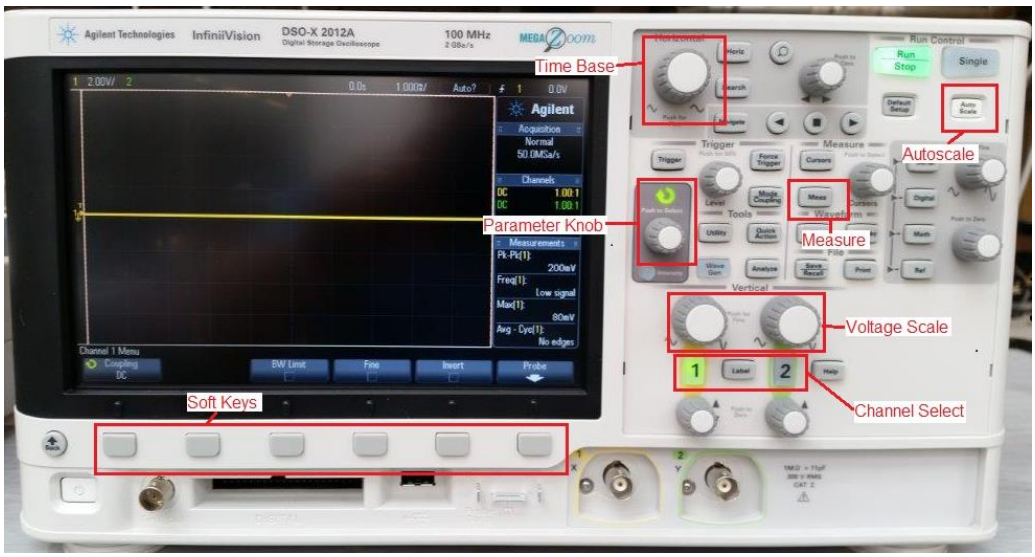

Figure 2: The interface of the Agilent DSO-X-2012A. Other scopes are similar.

Connect a probe to Channel 1 of the oscilloscope using the circular BNC connector. BNC connectors are pushed in and twisted slightly clockwise to lock. The other end of the probe has signal and ground separated.

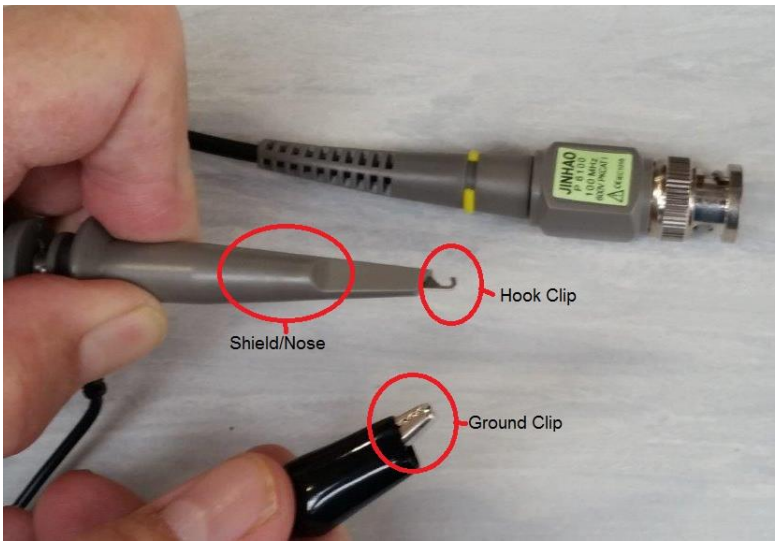

Figure 3: The ends of an oscilloscope probe.

3.1 PWM Waveforms and Measurements

First, find the schematic for the RCAP on Learn and open it. This will show you the layout for the two long connectors that link the Nucleo to the RCAP, as well as how all the peripherals on the RCAP are configured. Then:

1. Connect the ground lead of the probe to one of the corners marked `GND` on the Nucleo.
2. Connect the probe itself to pin 33 on `CN7` (somewhere on the top-left of the RCAP). Before you continue, try to work out which LED the probe is connected to using only the schematic; this will be a very useful skill in the weeks to come!
3. Press "Autoscale" ("Autoset" in the ESL) on the oscilloscope to quickly display the waveform.

The "1" button above the oscilloscope input should be glowing green, indicating that Channel 1 is active. In contrast, the "2" button should not be illuminated. Push each button a couple of times to individually activate and deactivate the different channels. You can control the voltage (vertical) and time (horizontal) scaling on the oscilloscope display by turning the relevant knobs, as shown in Figure 2. While they may not always be in the same place, these interfaces are common across all oscilloscopes.

You can also measure various properties of the waveform using the "Measurement Type" menu. To do this:

1. Press the "Meas" button on the oscilloscope and make sure that Source 1 is selected by pressing the far-left soft key. If the label above it does not say "Source 1", press the soft key repeatedly to change between available sources.
2. Select the property to be measured by pushing the second soft key from the left. This brings up a list of options, shown in Figure 4.

3. Cycle through these either by pressing the soft key again or by turning the parameter knob. Peak-to-peak voltage ("Pk-Pk") is a good choice.

Note that on the scopes in the ESL, the soft keys are replaced by a touch screen.

Figure 4: Soft keys to select options on the oscilloscope screen.

Make sure that the measurements you get match what you expect to see - never blindly assume that they are correct! In particular, check that your probe settings have the correct probe attenuation settings. Most oscilloscope probes have the option to change their attenuation, with $\times 1$ and $\times 10$ being the most common options. A $\times 10$ probe has 10 times the impedance of a $\times 1$ probe. This is useful when measuring high frequency signals, or large voltages, but a side-effect is that the measured voltage is 10 times smaller. We can account for this on the oscilloscope.

Figure 5: A passive alligator clip to BNC cable.

When using a more direct connection (e.g. that shown in Figure 5), we have a $\times 1$ connection (the cable is passive), but if the oscilloscope channel is set to $\times 10$ attenuation our measurement will be off by a factor of 10! You can check, and change, the attenuation setting on the oscilloscope by pressing the channel number, then press the soft key underneath the "Probe" label. Select the "Probe" label again, and use the parameter dial to select the desired probe attenuation.

3.2 Analogue Voltage

There is a potentiometer in the top-right corner of the RCAP. By looking at the schematic, work out which connector pin it is connected to. Using a female-to-male jumper wire, connect the probe to this pin. Hit the "Autoscale" button again and you should see a noisy, steady value. Twist the "Time Base" knob on the scope until it reaches "1 s/", or "one second per division". Now fiddle with the potentiometer. You should see the scope trace out the value slowly over time.

3.3 Digital Communications

Move the scope probe to pin 26 on `CN10`. Pressing autoscale here will likely result in a strobing mess; you'll need to manually change the amplitude and timebase knobs so that the blip fits neatly on the oscilloscope's screen. Once it fits, hit the "Single" button. This will make the solenoid stop trying to follow the signal continuously, but record a single instance and then pause. You can then zoom in on the signal to see the using those same knobs to look at the waveform. While you're not expected to decode this waveform, hopefully it looks familiar to something from ENCE260. If not, ask a friend or tutor to jog your memory.

4. STM32CubeIDE

Now we can test programs to make sure they are doing what we intended, let's make one ourselves. In this course we'll be using STM32CubeIDE to write and debug our code. Manufacturer-provided software is becoming less common in embedded systems thanks

to editors like VSCode, but is still in widespread use in industry. Most use the same open-source backend, and so learning how to drive one IDE will let you move between them easily.

Follow the "Introduction to STM32CubeIDE" on Learn to create a bare minimum project with some LEDs initialised. Next, open `main.c`. You will notice a lot of comments in here which are a bit daunting, but are really just telling you where you can and cannot add code. The reason for this is that the configurator tool needs to overwrite code when you change the settings in the `.ioc` file, and so only some areas of the generated files are safe. We'll discuss a way to get around this next lab, but for now navigate to the `while (1)` loop and add the following two lines of code as shown:

```
  /* Infinite loop */
  /* USER CODE BEGIN WHILE */
  while (1)
  {
    /* USER CODE END WHILE */

    /* USER CODE BEGIN 3 */
      HAL_GPIO_TogglePin(LD1_GPIO_Port, LD1_Pin);
      HAL_Delay(100);
  }
  /* USER CODE END 3 */
}

/**
  * @brief System Clock Configuration
  * @retval None
```

Figure 6: Code additions.

Make sure your board is connected to the computer via the correct USB port as described above, then click the green play circle in the top toolbar. If a window appears asking you to configure the debugger settings, just click "OK". You should now see a green LED near the USB on the board flashing!

After admiring your work, probe pin 11 on `CN10` and have a look at the waveform. Then, increase the flash rate in software to 100 Hz, and measure it using the scope's measurement menu. Confirm that the frequency the scope measures is the frequency you expect.

5. Final Tasks

Leave the boards on the desks where you found them, turn off the scope, and pack away any leads. **Make sure you join a group on Learn before the end of Thursday, otherwise you will have no control over who your partner is.** We'll move beyond just blinking an LED next week.